

## GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>Project</b>                 | FEED for Gas Gathering Pipelines & Associated Infrastructure |
| <b>Client</b>                  | Brass Fertilizer and Petrochemical Company Limited           |
| <b>Originating Company</b>     | Epic Atlantic Limited                                        |
| <b>Document Title</b>          | <b>Specification Pressure Safety Valve</b>                   |
| <b>Document Number</b>         | EA-BFPCL-GPMC-GGPL-FEED- ICS-DSH-039                         |
| <b>Document Revision</b>       | R02                                                          |
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| <b>Originator</b>              | Paul Roy                                                     |
| <b>Security Classification</b> | Restricted                                                   |
| <b>Issue Date</b>              | <b>19-Oct-22</b>                                             |

## Specification for Pressure Safety Valve

|       |                   |                           |                   |                 |                        |                         |
|-------|-------------------|---------------------------|-------------------|-----------------|------------------------|-------------------------|
|       |                   |                           |                   |                 |                        |                         |
| R02   | 19.10.22          | Re-Issued for Review      | P. Roy            | P. Olukah       | A. Duru                |                         |
| R01   | 18.07.22          | Issued for Review         | P. Roy            | P. Olukah       | A. Duru                |                         |
| Rev # | <b>Issue Date</b> | <b>Status Description</b> | <b>Originator</b> | <b>Reviewer</b> | <b>Approver (Epic)</b> | <b>Approver (BFPCL)</b> |

## Revision History

| Rev No. | Date     | Description of Revision                               |
|---------|----------|-------------------------------------------------------|
| R01     | 18.07.22 | Issued for Review                                     |
| R02     | 19.10.22 | Re-issued for review after incorporating TCE comments |
|         |          |                                                       |
|         |          |                                                       |

## Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)  
*for the subsequent project phases, post FEED, it is expected that:*
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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## SUMMARY PAGE

|                  |                                                     |                                                        |
|------------------|-----------------------------------------------------|--------------------------------------------------------|
| <b>Client:</b>   | Brass Fertilizer & Petrochemical Company Limited    |                                                        |
| <b>Project:</b>  | Gas Processing & Methanol Complex                   | <b>Project No.</b> 11998A                              |
| <b>Facility:</b> | Gas Gathering Pipelines & Associated Infrastructure | <b>Doc No.</b><br>EA-BFPCL-GPMC-GGPL-FEED- ICS-DSH-039 |

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

## REFERENCES

| Document No.                          | Description                                                 |
|---------------------------------------|-------------------------------------------------------------|
| EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001   | Upstream Gas Gathering Concept Select & Revalidation Report |
| EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001 | Preliminary Simulation – Based on Desktop Route Survey      |
| EA/SPDC-BFPC UGS-GEP/CSM-RPT-002      | Gas Gathering & Evacuation Pipelines Concept Screening      |
| EA/SPDC-BFPC UGS-GEP/CSM-RPT-001      | Conceptual Study Memorandum                                 |
| EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002   | Basis of Design                                             |
|                                       | FISAS Draft ESIA Report (November 2020)                     |
|                                       |                                                             |

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# 1. Introduction

## 1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

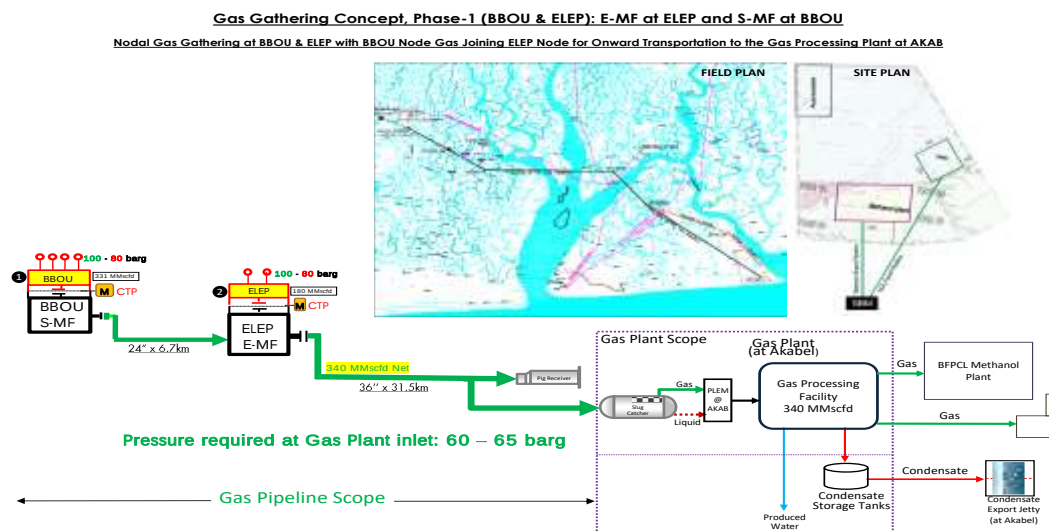


Figure 1: Gas Gathering Concept, Phase 1

## 1.2. Purpose

This specification covers the minimum requirements for the design, manufacture, assembly, inspection, testing, certification, delivery for Pressure Safety Valves for BBOU, ELEM and AKAB.

The FEED shall fully comply with all specified relevant contractual requirements.

## 2. Regulations, Codes And Standards

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The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to.

The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

### 3. General

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- 3.1.** The vendor shall be responsible for verification of the selection of valve size, type and material for each application in accordance with the information furnished in this document.
- 3.2.** A permanently attached (with drive pins or equal) stainless steel tag (s) stamped with the following information shall be provided on each valve.
- Tag Number
  - Manufacturer's Name or Trade Mark
  - Model Number
  - Serial Number
  - Size of Valve with flange rating for inlet and outlet
  - Body and Trim Materials
  - Set Pressure
  - Orifice Description
  - CODE STAMP

## 4. Safety and Relief Systems

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### 4.1. Safety Relief valves

- 4.1.1. Safety and relief valves shall meet the requirements of the ASME unfired pressure vessel code, API RP 520, and API RP 521.
- 4.1.2. Safety Valves for BBOU, ELEM and AKAB shall be designed as per 'External Fire' case.
- 4.1.3. Safety Valve for BBOU, ELEM and AKAB shall be 'conventional' type.
- 4.1.4. Safety relief valve orifice sizes shall be calculated on a pure 'area' requirement basis, capacity adjustments required by code revisions or other applicable regulations shall be covered within valve selection procedure.
- 4.1.5. Safety valves for general process service shall be direct spring loaded with high lift characteristics and be flanged with minimum carbon-steel body and stainless-steel nozzle, disc. and guides.
- 4.1.6. Body and Bonnet of the Safety Valve shall A216 WCB.
- 4.1.7. Nozzle, disc and spring shall be SS316
- 4.1.8. Vendor shall size the safety valve as per the process data given herein this document.



## 5. Design & Construction

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- 5.1. Trim shall include the nozzle, the disc holder, stem, guide, blowdown ring, ring pin, bushing and all internal parts, except the body, in contact with the process fluid. The nozzle and disc shall be Forged / bar stock SS 316 with machined stainless-steel guide and spindle.
- 5.2. Caps shall be provided over the spring adjusting screws on all valves except as required by codes. The cap and bonnet shall be provided with a wire seal, unless otherwise specified, to prevent unauthorised adjustment.
- 5.3. Only safety relief valves incorporating adjustable blowdown rings shall be offered for protecting process equipment. In liquid service the blowdown ring shall be made inoperative by the manufacturers.

## 6. Functional Requirements

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- 6.1. The safety or relief valve shall be of adequate size to relieve the capacity at the specified pressure and temperature conditions and fluid physical properties regardless of any preliminary sizing calculations or orifice designation.
- 6.2. Safety valve shall be sized as per the External Fire Case with allowable over pressure of 21% and Blowdown maximum 10%.
- 6.3. A capacity correction shall be made if the back pressure exceeds the critical pressure drop for vapours and gases.
- 6.4. The valves shall be suitable for continuous operation under the operating conditions indicated in data sheets.
- 6.5. For very low capacity applications, if API orifice is offered, vendor shall provide mechanical stop to prevent lifting. Vendor shall also arrange for witnessing capacity test of such valves
- 6.6. Valve spring shall be selected suitably to permit  $\pm 5\%$  min. adjustment of the set pressure.
- 6.7. Pressure relief valves shall be full nozzle full lift type.
- 6.8. All Pressure Relief Valves shall be type tested for capacity as per ASME Sec VIII UG-131 and the actual blowdown of individual valves shall meet the process requirement. Vendor shall submit the type test certificate.

## 7. Performance

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- 7.1. Each valve shall be factory set and tested in accordance with the appropriate code procedures.
- 7.2. When the valve is subjected to the specified set pressure, destructive chattering shall not occur when raising the pressure to the popping point and then lowering to below the closing pressure.
- 7.3. Set pressure tolerances shall be as required by the appropriate code.
- 7.4. Operational tightness test - the valves shall be bubble tight when tested at 90% of set pressure in accordance with the appropriate standard
- 7.5. Inspection and Testing- Radiography, Hydrostatic, PMI etc. shall be considered as per approved Quality Assurance Plan.

## 8. Codes

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Unless otherwise specified the design shall conform to the requirement of:

|                         |  |                                                                                                                                                                                  |
|-------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API RP 520              |  | Design and Installation of Pressure Relieving Systems in Refineries, Part I. Sizing selection and installation of pressure relieving devices in refineries Part II installation. |
| API Std 526             |  | Flanged Steel Safety Relief Valves for use in Petroleum Refineries                                                                                                               |
| API Std 527             |  | Commercial Seat Tightness of Safety Valves with Metal to Metal Seats.                                                                                                            |
| ASME Section VIII Div I |  | Boiler and Pressure Vessel Code.                                                                                                                                                 |

Unless otherwise specified latest edition of codes to be followed.

## 9. Flow Data

### Production Forecast & Fluid Characteristics

Notional Fields Forecasts of Well Head Gas (WHG) Supply Required For Export Of 340 MMSCFPD Of Lean Gas To Methanol Plant For 20 Years With Elepa Development Accelerated to Year 1 OSD with Bubouwe Bou

#### BUBOUWE BOU

Condensate-Gas Ratio (CGR) is 5.56 BBL/MMscf.

| Time (Year) | Gas (MMSCF/D) | Condensate (BCPD) | Water (BWPD) |
|-------------|---------------|-------------------|--------------|
| 1           | 181           | 944               | 907          |
| 2           | 181           | 878               | 907          |
| 3           | 181           | 788               | 907          |
| 4           | 181           | 742               | 7855         |
| 5           | 181           | 707               | 7547         |
| 6           | 215           | 840               | 8491         |
| 7           | 305           | 1217              | 9002         |
| 8           | 327           | 1323              | 8193         |
| 9           | 331           | 1338              | 8038         |
| 10          | 67            | 239               | 2397         |
| 11          | 21            | 59                | 1031         |
| 12          | 16            | 42                | 784          |
| 13          | 2             | 4                 | 85           |
| 14          | 0             | 0                 | 0            |
| 15          | 0             | 0                 | 0            |
| 16          | 0             | 0                 | 0            |
| 17          | 0             | 0                 | 0            |
| 18          | 0             | 0                 | 0            |
| 19          | 0             | 0                 | 0            |

|     |        |         |         |
|-----|--------|---------|---------|
| 20  | 0      | 0       | 0       |
| TCQ | 800    | 3       | 21      |
|     | (BSCF) | (MMBBL) | (MMBBL) |

**9.1.**

ELEPA

Condensate-Gas Ratio (CGR) is 60.64 BBL/MMscf.

| Time<br>(Year) | Gas<br>(MMSCF/D) | Condensate<br>(BCPD) | Water<br>(BWPD) |
|----------------|------------------|----------------------|-----------------|
| 1              | 180              | 10088                | 900             |
| 2              | 180              | 10088                | 900             |
| 3              | 180              | 10088                | 900             |
| 4              | 180              | 10088                | 7980            |
| 5              | 180              | 10088                | 7666            |
| 6              | 145              | 8146                 | 4902            |
| 7              | 52               | 2911                 | 2597            |
| 8              | 29               | 1630                 | 1454            |
| 9              | 25               | 1386                 | 1236            |
| 10             | 19               | 1076                 | 960             |
| 11             | 11               | 628                  | 561             |
| 12             | 9                | 478                  | 426             |
| 13             | 1                | 52                   | 46              |
| 14             | 0                | 0                    | 0               |
| 15             | 0                | 0                    | 0               |
| 16             | 0                | 0                    | 0               |
| 17             | 0                | 0                    | 0               |
| 18             | 0                | 0                    | 0               |
| 19             | 0                | 0                    | 0               |
| 20             | 0                | 0                    | 0               |

|     |        |         |         |
|-----|--------|---------|---------|
| TCQ | 435    | 24      | 11      |
|     | (BSCF) | (MMBBL) | (MMBBL) |

## NUN RIVER

Condensate-Gas Ratio (CGR) is 67.57 BBL / MMscf.

| Time<br>(Year) | Gas<br>(MMSCF/D) | Condensate<br>(BCPD) | Water<br>(BWPD) |
|----------------|------------------|----------------------|-----------------|
| 1              | 0                | 0                    | 0               |
| 2              | 0                | 0                    | 0               |
| 3              | 0                | 0                    | 0               |
| 4              | 0                | 0                    | 0               |
| 5              | 0                | 0                    | 0               |
| 6              | 0                | 0                    | 0               |
| 7              | 0                | 0                    | 0               |
| 8              | 0                | 0                    | 0               |
| 9              | 0                | 0                    | 0               |
| 10             | 244              | 14464                | 1221            |
| 11             | 245              | 14583                | 1225            |
| 12             | 245              | 14583                | 1225            |
| 13             | 245              | 14583                | 10844           |
| 14             | 245              | 14583                | 10417           |
| 15             | 196              | 10914                | 6653            |
| 16             | 71               | 4159                 | 3528            |
| 17             | 40               | 2329                 | 1976            |
| 18             | 34               | 1980                 | 1680            |
| 19             | 26               | 1538                 | 1304            |
| 20             | 15               | 898                  | 762             |
| TCQ            | 586              | 35                   | 15              |
|                | (BSCF)           | (MMBBL)              | (MMBBL)         |

SEIB+OPUG+ADD\*\*

Condensate-Gas Ratio (CGR) is 65.63 BBL/MMscf.

| Time<br>(Year) | Gas<br>(MMSCF/D) | Condensate<br>(BCPD) | Water<br>(BWPD) |
|----------------|------------------|----------------------|-----------------|
| 1              | 0                | 0                    | 0               |
| 2              | 0                | 0                    | 0               |
| 3              | 0                | 0                    | 0               |
| 4              | 0                | 0                    | 0               |
| 5              | 0                | 0                    | 0               |
| 6              | 0                | 0                    | 0               |
| 7              | 0                | 0                    | 0               |
| 8              | 0                | 0                    | 0               |
| 9              | 0                | 0                    | 0               |
| 10             | 52               | 3049                 | 259             |
| 11             | 108              | 6367                 | 541             |
| 12             | 116              | 6831                 | 580             |
| 13             | 139              | 8161                 | 2857            |
| 14             | 141              | 8322                 | 6096            |
| 15             | 188              | 11082                | 7496            |
| 16             | 308              | 18164                | 7717            |
| 17             | 341              | 24229                | 6297            |
| 18             | 348              | 32453                | 3334            |
| 19             | 354              | 31623                | 3116            |
| 20             | 363              | 31213                | 5551            |
| TCQ            | 898              | 66                   | 16              |
|                | (BSCF)           | (MMBBL)              | (MMBBL)         |

## TOTAL SUPPLY PORTFOLIO

|      |                          |            |                                |
|------|--------------------------|------------|--------------------------------|
|      | <b>Total WHG Supply*</b> |            | <b>CGR (BBL/MMscf) = 64.14</b> |
| Time | GAS                      | CONDENSATE | WATER                          |



|      | Total WHG Supply* |            | CGR (BBL/MMscf) = 64.14 |
|------|-------------------|------------|-------------------------|
| year | MMSCF/D           | BCPD       | BWPD                    |
| Year | Gas               | Condensate | Water                   |
| 1    | 361               | 11032      | 1807                    |
| 2    | 361               | 10965      | 1807                    |
| 3    | 361               | 10876      | 1807                    |
| 4    | 361               | 10830      | 15835                   |
| 5    | 361               | 10794      | 15213                   |
| 6    | 360               | 8985       | 13393                   |
| 7    | 357               | 4128       | 11599                   |
| 8    | 356               | 2953       | 9647                    |
| 9    | 356               | 2724       | 9275                    |
| 10   | 382               | 18828      | 4837                    |
| 11   | 385               | 21637      | 3357                    |
| 12   | 385               | 21933      | 3015                    |
| 13   | 386               | 22799      | 13832                   |
| 14   | 386               | 22905      | 16513                   |
| 15   | 384               | 21996      | 14149                   |
| 16   | 379               | 22323      | 11246                   |
| 17   | 380               | 26558      | 8272                    |
| 18   | 382               | 34433      | 5014                    |
| 19   | 380               | 33161      | 4421                    |
| 20   | 379               | 32111      | 6313                    |
|      | 2719              | 129        | 63                      |
|      | (BSCF)            | (MMBBL)    | (MMBBL)                 |

## Notes

WHG Supply\* on Operating Day basis AND NOT Calendar day basis

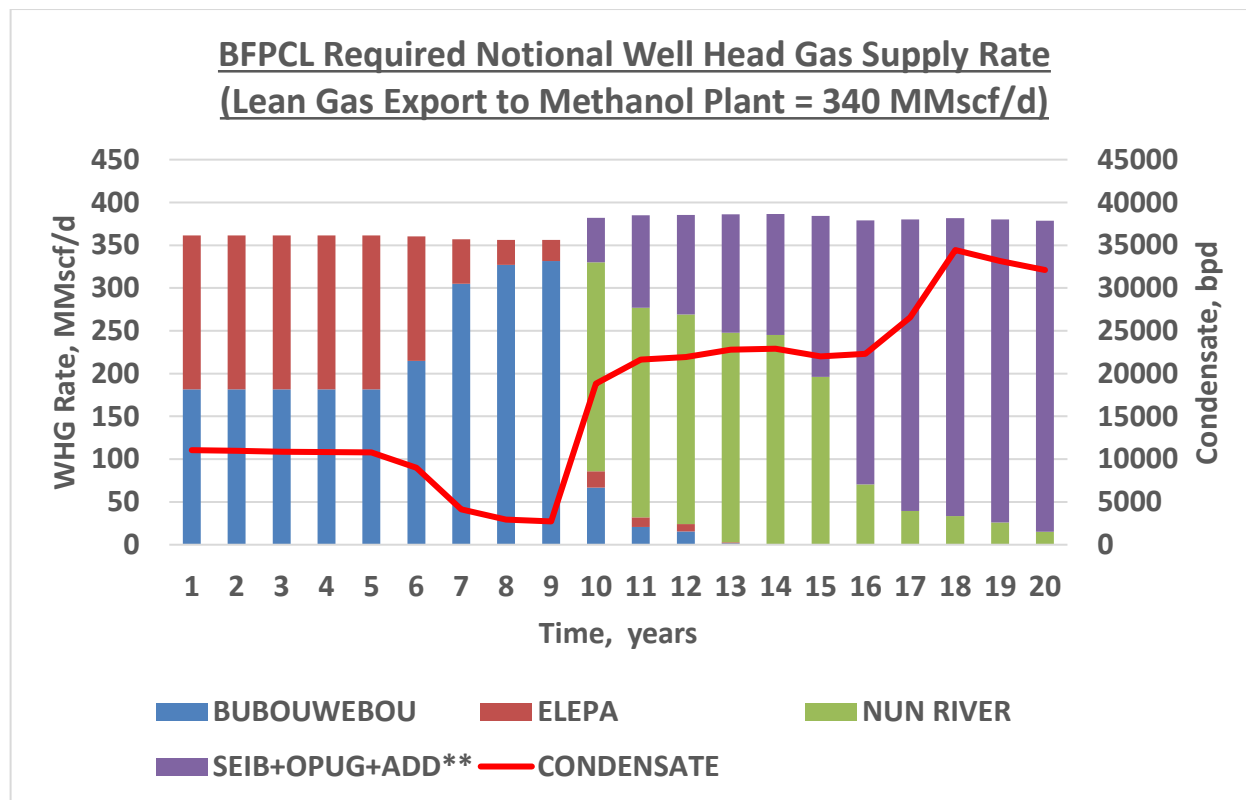
SEIB+OPUG+ADD\*\* is subject to SPDC providing requested information

Delivery Pressure range at Custody Transfer Point: 80 - 100 bar

Delivery Temperature range at Custody Transfer Point: 50 - 95 degC

## BFPC-L Required Notional Well Head Gas Supply Rate

(Lean Gas Export to Methanol Plant = 340 MMscf/d)



## Gas Compositions from Nominated Fields

## Field Compositions

| Field 1: Bubouwe Bou Analogue |       | Field 2: Elepa |       | Field 3: Nun River H4.0 |       |
|-------------------------------|-------|----------------|-------|-------------------------|-------|
| Component                     | Mol.% | Component      | Mol.% | Component               | Mol.% |
| N2                            | 0.07  | N2             | 0     | N2                      | NA    |
| CO2                           | 0.59  | CO2            | 2.22  | CO2                     | 0.77  |
| C1                            | 92.92 | C1             | 81.66 | C1                      | 72.83 |
| C2                            | 3.16  | C2             | 6.13  | C2                      | 7.68  |
| C3                            | 1.01  | C3             | 2.59  | C3                      | 4.85  |
| iC4                           | 0.61  | iC4            | 0.75  | iC4                     | 1.46  |
| nC4                           | 0.44  | nC4            | 0.81  | nC4                     | 1.97  |
| iC5                           | 0.25  | iC5            | 0.45  | iC5                     | 0.95  |
| nC5                           | 0.19  | nC5            | 0.33  | nC5                     | 0.76  |
| C6                            | 0.23  | C6             | 0.65  | C6                      | 1.03  |

| Field1: Bubouwe Bou Analogue |      | Field 2: Elepa |      | Field 3: Nun River H4.0 |       |
|------------------------------|------|----------------|------|-------------------------|-------|
| C7                           | 0.19 | C7             | 0.9  | C7                      | 1.41  |
| C8                           | 0.04 | C8             | 0.87 | C8                      | 1.61  |
| C9                           | 0.03 | C9             | 0.38 | C9                      | 0.9   |
| C10                          | 0.02 | C10            | 0.3  | C10                     | 0.65  |
| C11                          | 0.03 | C11            | 0.23 | C11                     | 0.47  |
| C12+                         | 0.22 | C12            | 0.22 | C12                     | 0.37  |
| C13                          | NA   | C13            | 0.23 | C13                     | 0.34  |
| C14                          | NA   | C14            | 0.22 | C14                     | 0.3   |
| C15                          | NA   | C15            | 0.19 | C15                     | 0.26  |
| C16                          | NA   | C16            | 0.15 | C16                     | 0.21  |
| C17                          | NA   | C17            | 0.12 | C17                     | 0.16  |
| C18                          | NA   | C18            | 0.11 | C18                     | 0.14  |
| C19                          | NA   | C19            | 0.06 | C19                     | 0.08  |
| C20+                         | NA   | C20+           | 0.43 | C20+                    | 0.8   |
| TOTAL                        | 100  | TOTAL          | 100  | TOTAL                   | 100   |
| Flash CGR                    | 7    | Flash CGR      | 59.7 | Flash CGR               | 161.7 |

## Portfolio Composition

| Range for Portfolio | Lean Sample |  | Rich Sample |
|---------------------|-------------|--|-------------|
| Component           | Mol.%       |  | Mol.%       |
| N2                  | 0.07        |  | NA          |
| CO2                 | 0.59        |  | 0.77        |
| C1                  | 92.92       |  | 72.83       |
| C2                  | 3.16        |  | 7.68        |
| C3                  | 1.01        |  | 4.85        |
| iC4                 | 0.61        |  | 1.46        |
| nC4                 | 0.44        |  | 1.97        |

| Range for Portfolio | Lean Sample |  | Rich Sample |
|---------------------|-------------|--|-------------|
| iC5                 | 0.25        |  | 0.95        |
| nC5                 | 0.19        |  | 0.76        |
| C6                  | 0.23        |  | 1.03        |
| C7                  | 0.19        |  | 1.41        |
| C8                  | 0.04        |  | 1.61        |
| C9                  | 0.03        |  | 0.9         |
| C10                 | 0.02        |  | 0.65        |
| C11                 | 0.03        |  | 0.47        |
| C12+                | 0.22        |  | 0.37        |
| C13                 | NA          |  | 0.34        |
| C14                 | NA          |  | 0.3         |
| C15                 | NA          |  | 0.26        |
| C16                 | NA          |  | 0.21        |
| C17                 | NA          |  | 0.16        |
| C18                 | NA          |  | 0.14        |
| C19                 | NA          |  | 0.08        |
| C20+                | NA          |  | 0.8         |
| TOTAL               | 100         |  | 100         |

## Notes

STB - Stock Tank Barrel

MMscf - Million standard cubic feet

Well Compositions

BBOU D3

As used in simulation model - Tuned PVTsim Model using Bonn-31ST PVT as analogue.

| Component | Mol %  | Molecular Weight | Liquid Density (g/cc) |
|-----------|--------|------------------|-----------------------|
| N2        | 0.070  | 28.014           |                       |
| CO2       | 0.590  | 44.010           |                       |
| C1        | 92.922 | 16.043           |                       |

| Component | Mol %   | Molecular Weight | Liquid Density (g/cc) |
|-----------|---------|------------------|-----------------------|
| C2        | 3.160   | 30.070           |                       |
| C3        | 1.010   | 44.097           |                       |
| iC4       | 0.610   | 58.124           |                       |
| nC4       | 0.440   | 58.124           |                       |
| iC5       | 0.250   | 72.151           |                       |
| nC5       | 0.190   | 72.151           |                       |
| C6        | 0.230   | 86.178           | 0.664                 |
| C7        | 0.190   | 96.000           | 0.738                 |
| C8        | 0.040   | 107.000          | 0.765                 |
| C9        | 0.030   | 121.000          | 0.781                 |
| C10-C11   | 0.050   | 141.800          | 0.794                 |
| C12       | 0.068   | 161.000          | 0.800                 |
| C13       | 0.047   | 175.000          | 0.805                 |
| C14       | 0.032   | 190.000          | 0.808                 |
| C15       | 0.022   | 206.000          | 0.812                 |
| C16       | 0.015   | 222.000          | 0.815                 |
| C17       | 0.010   | 237.000          | 0.818                 |
| C18       | 0.007   | 251.000          | 0.821                 |
| C19-C44   | 0.016   | 293.067          | 0.830                 |
| Total     | 100.000 |                  |                       |

| Bonny 31ST Reservoir Fluid Composition |           |           |           |                 |
|----------------------------------------|-----------|-----------|-----------|-----------------|
| No                                     | Component | Flash Gas | Flash Oil | Reservoir Fluid |
|                                        |           | [mol %]   | [mol %]   | [mol %]         |
| 1                                      | N2        | 0.07      | 0         | 0.07            |
| 2                                      | CO2       | 0.59      | 0         | 0.59            |
| 3                                      | C1        | 93.26     | 0         | 92.92           |
| 4                                      | C2        | 3.17      | 0.01      | 3.16            |

|    |      |      |       |      |
|----|------|------|-------|------|
| 5  | C3   | 1.01 | 0.03  | 1.01 |
| 6  | i-C4 | 0.61 | 0.07  | 0.61 |
| 7  | n-C4 | 0.44 | 0.07  | 0.44 |
| 8  | i-C5 | 0.25 | 0.16  | 0.25 |
| 9  | n-C5 | 0.19 | 0.16  | 0.19 |
| 10 | C6   | 0.22 | 1.88  | 0.23 |
| 11 | C7   | 0.17 | 6.21  | 0.19 |
| 12 | C8   | 0.02 | 6.61  | 0.04 |
| 13 | C9   | 0    | 7.13  | 0.03 |
| 14 | C10  | 0    | 5.7   | 0.02 |
| 15 | C11  | 0    | 7.83  | 0.03 |
| 16 | C12+ | 0    | 64.14 | 0.22 |

|                   |           |        |        |            |
|-------------------|-----------|--------|--------|------------|
| Total             |           | 100    | 100    | 100        |
| Density @ 60F     | [g/cm^3]  |        | 0.811  |            |
| MW                | [g/mol]   | 17.97  | 165.61 | 18.47      |
| GOR               | [Scf/Stb] |        |        | 180,011.90 |
| Gravity           | [air=1]   | 0.62   |        |            |
| C7+               | Mole %    | 0.45   | 97.62  | 0.53       |
| C7+ MW            | [g/mol]   | 100.89 | 167.69 | 143.17     |
| C12+ MW           | [g/mol]   | -      | 191.6  | 191.6      |
| CGR (bbls/mm scf) |           |        |        | 5.56       |

## Elepa U-4

| Component | Res Fluid | Liquid |      | Gas |      |
|-----------|-----------|--------|------|-----|------|
|           | (z)       |        | (x)  |     | (y)  |
| Totals    | mol%      |        | mol% |     | mol% |
| N2        | 0.0000    |        | 0.00 |     | 0.00 |
| CO2       | 2.2200    |        | 0.00 |     | 2.30 |
| H2S       | 0.0000    |        | 0.00 |     | 0.00 |

| Component | Res Fluid | Liquid |        | Gas |        |
|-----------|-----------|--------|--------|-----|--------|
| c1        | 81.6600   |        | 0.00   |     | 84.63  |
| c2        | 6.1300    |        | 0.03   |     | 6.35   |
| c3        | 2.5900    |        | 0.27   |     | 2.67   |
| ic4       | 0.7500    |        | 0.37   |     | 0.76   |
| nc4       | 0.8100    |        | 0.54   |     | 0.82   |
| ic5       | 0.4500    |        | 0.87   |     | 0.43   |
| nc5       | 0.3300    |        | 0.68   |     | 0.32   |
| c6        | 0.6500    |        | 2.26   |     | 0.59   |
| c7        | 0.9000    |        | 6.75   |     | 0.69   |
| c8        | 0.8700    |        | 12.76  |     | 0.44   |
| c9        | 0.3800    |        | 10.87  |     | 0.00   |
| c10       | 0.3000    |        | 8.62   |     | 0.00   |
| c11       | 0.2300    |        | 6.64   |     | 0.00   |
| c12       | 0.2200    |        | 6.15   |     | 0.00   |
| c13       | 0.2300    |        | 6.45   |     | 0.00   |
| c14       | 0.2200    |        | 6.16   |     | 0.00   |
| c15       | 0.1900    |        | 5.51   |     | 0.00   |
| c16       | 0.1500    |        | 4.37   |     | 0.00   |
| c17       | 0.1200    |        | 3.36   |     | 0.00   |
| c18       | 0.1100    |        | 3.02   |     | 0.00   |
| c19       | 0.0600    |        | 1.62   |     | 0.00   |
| c20+      | 0.4300    |        | 12.70  |     | 0.00   |
| Totals    | 100.00    |        | 100.00 |     | 100.00 |

## Elepa U-7

| Component | Res Fluid | Liquid |      | Gas |      |
|-----------|-----------|--------|------|-----|------|
|           | (z)       |        | (x)  |     | (y)  |
| Totals    | mol%      |        | mol% |     | mol% |

| Component | Res Fluid | Liquid |        | Gas |        |
|-----------|-----------|--------|--------|-----|--------|
| co2       | 2.1400    |        | 0.00   |     | 2.16   |
| N         | 0.0000    |        | 0.00   |     | 0.00   |
| c1        | 86.6700   |        | 0.00   |     | 87.30  |
| c2        | 6.2100    |        | 0.37   |     | 6.25   |
| c3        | 2.0500    |        | 0.24   |     | 2.06   |
| ic4       | 0.5200    |        | 0.16   |     | 0.52   |
| nc4       | 0.4900    |        | 0.26   |     | 0.49   |
| ic5       | 0.2500    |        | 0.39   |     | 0.41   |
| nc5       | 0.1600    |        | 0.36   |     |        |
| c6        | 0.2800    |        | 1.88   |     | 0.27   |
| c7        | 0.3700    |        | 5.97   |     | 0.33   |
| c8        | 0.3100    |        | 13.52  |     | 0.21   |
| c9        | 0.1000    |        | 13.33  |     | 0.00   |
| c10       | 0.0700    |        | 10.03  |     | 0.00   |
| c11       | 0.0500    |        | 7.46   |     | 0.00   |
| c12       | 0.0500    |        | 6.75   |     | 0.00   |
| c13       | 0.0500    |        | 6.70   |     | 0.00   |
| c14       | 0.0400    |        | 6.01   |     | 0.00   |
| c15       | 0.0400    |        | 4.99   |     | 0.00   |
| c16       | 0.0300    |        | 4.02   |     | 0.00   |
| c17       | 0.0200    |        | 3.00   |     | 0.00   |
| c18       | 0.0200    |        | 2.84   |     | 0.00   |
| c19       | 0.0200    |        | 2.23   |     | 0.00   |
| c20+      | 0.0700    |        | 9.49   |     | 0.00   |
| Totals    | 100.01    |        | 100.00 |     | 100.00 |



## 10. Tag Nos.

| Sr. No. | Tag No.       | Size and Rating          | Orifice Designation | Set Pressure |
|---------|---------------|--------------------------|---------------------|--------------|
| 1       | BBOU-PSV-1001 | 4" 900# RTJ X 6" 150# RF | L                   | 110 barg     |
| 2       | BBOU-PSV-1002 | 4" 900# RTJ X 6" 150# RF | L                   | 110 barg     |
| 3       | BBOU-PSV-1003 | 4" 900# RTJ X 6" 150# RF | L                   | 110 barg     |
| 4       | ELEP-PSV-1001 | 4" 900# RTJ X 6" 150# RF | L                   | 110 barg     |
| 5       | ELEP-PSV-1002 | 4" 900# RTJ X 6" 150# RF | L                   | 110 barg     |
| 6       | ELEP-PSV-1003 | 4" 900# RTJ X 6" 150# RF | L                   | 110 barg     |
| 7       | ELEP-PSV-1004 | 4" 900# RTJ X 6" 150# RF | L                   | 110 barg     |
| 8       | AKAB-PSV-1001 | 4" 900# RTJ X 6" 150# RF | L                   | 110 barg     |

Tagwise datasheets including process parameters shall be provided during Detailed Engineering Design.